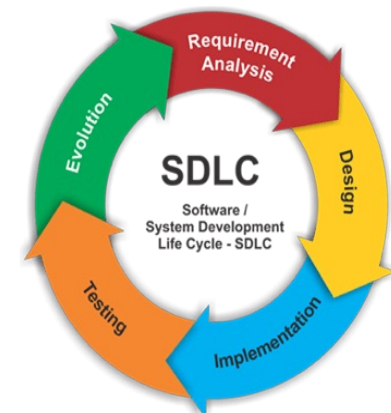
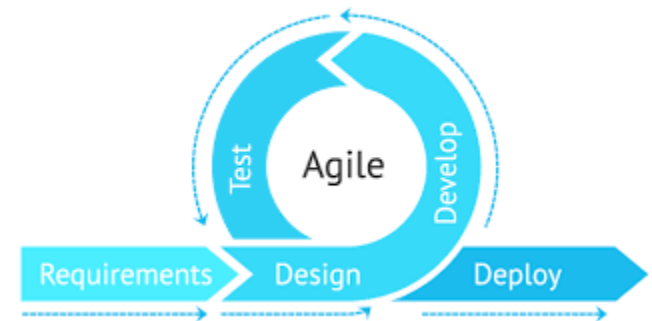


Chapter 2: Software Development Life Cycle - 2



Activity – Research about “Agile”

1. What is Agile?
2. What is Agile mindset?
3. Popular agile methods



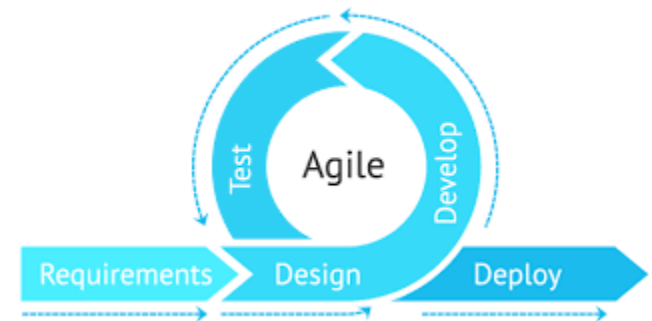
Agile Model



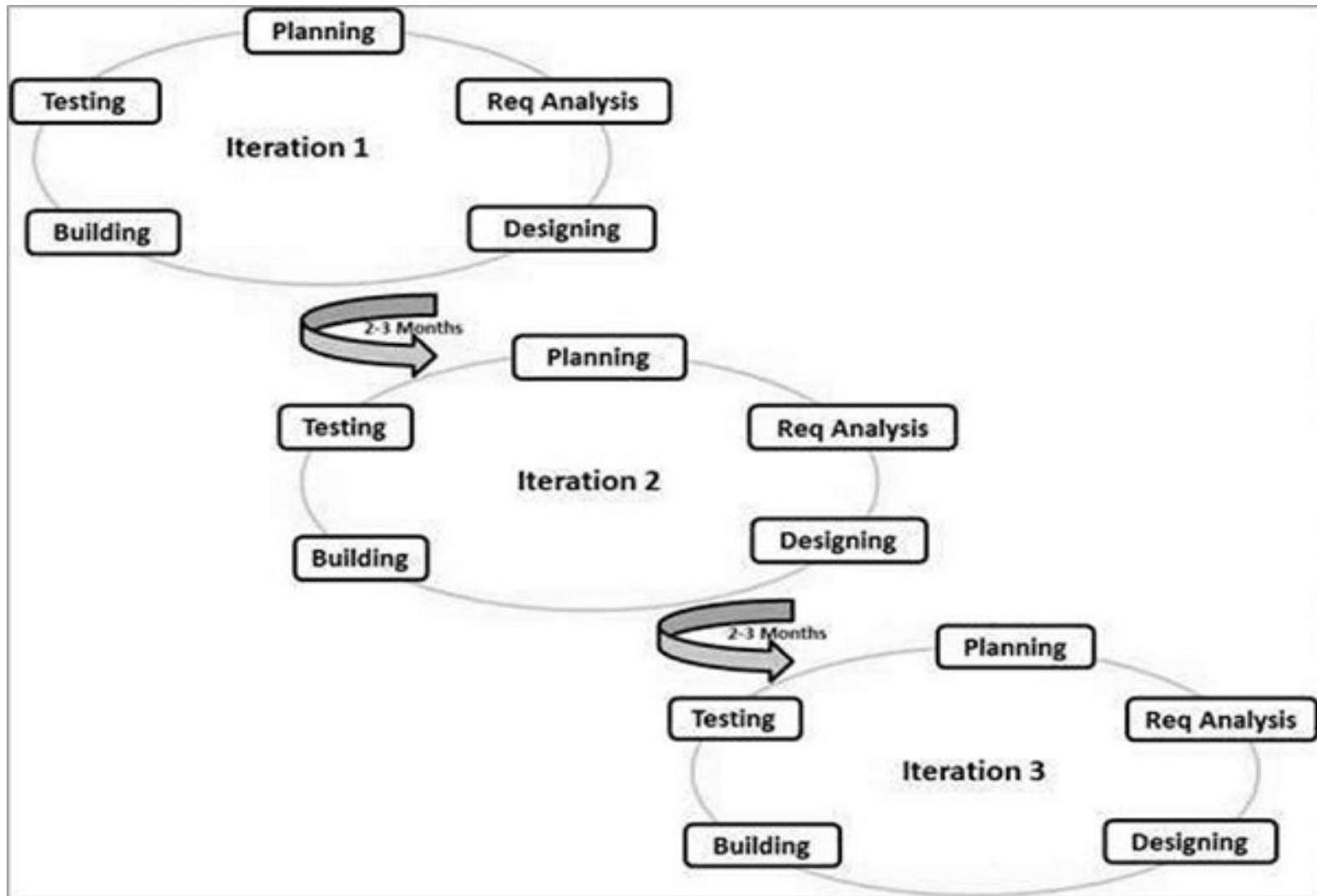
Agile Model

Agile SDLC model is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software product.

In Agile, both development and testing activities are done concurrently.



Agile Model



Agile Model

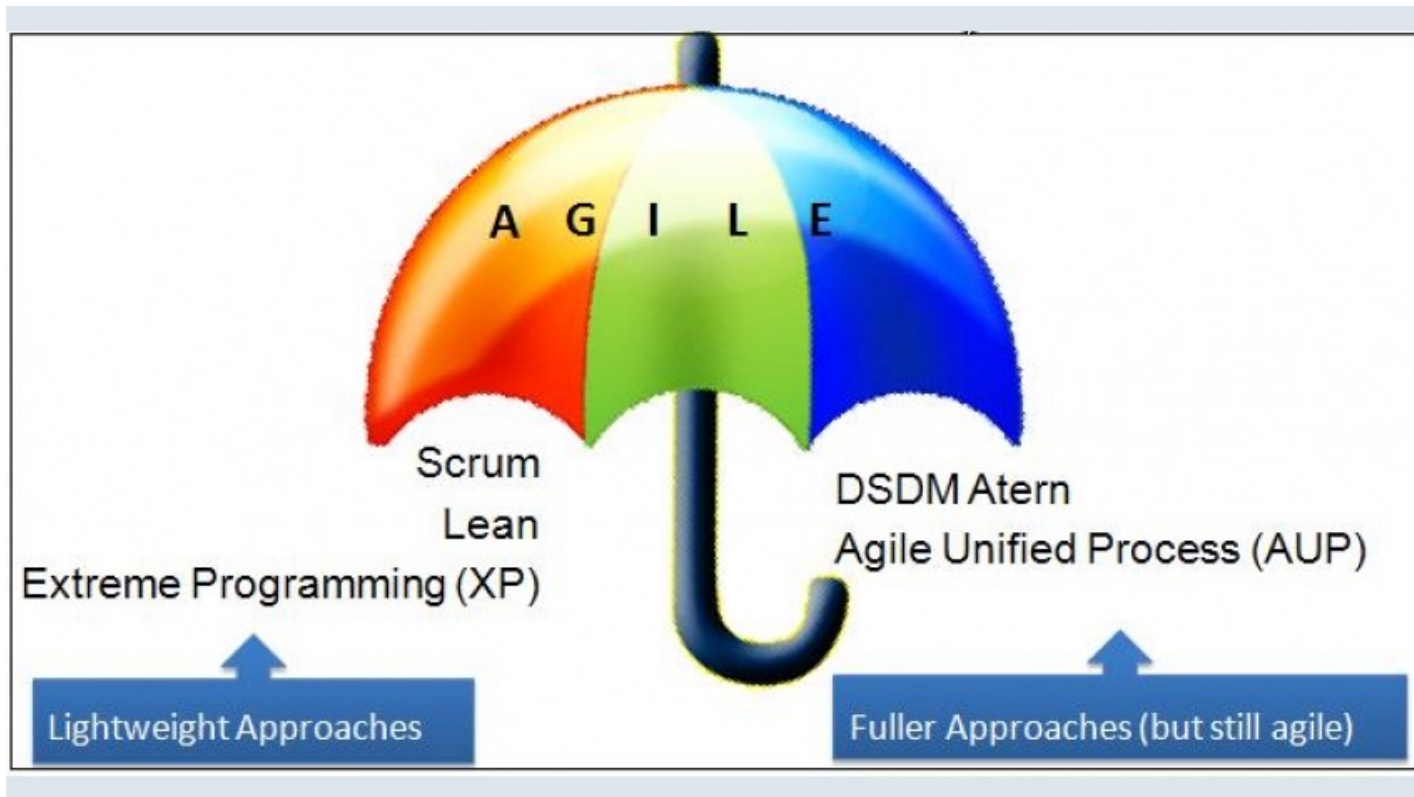
Agile Manifesto Principles -

- Individuals & Interaction.
- Customer Collaboration.
- Responding to changes.
- Working Software.
- Working together.
- Simplicity.



Agile Model

Popular Agile Methods



Agile Model

Advantages

- Easy to manage.
- Gives flexibility to developers.
- Good model for environments that change steadily.
- Minimal rules, documentation easily employed.
- Is a very realistic approach to software development
- Enables concurrent development and delivery within an overall planned context.



Agile Model

Disadvantages

- Agile model brought agility but it's lost on operations so there is **no speed in agile practices**.
- Considerable amount of skills & training are required for successful implementation.
- Depends heavily on customer interaction, so if customer is not clear, team can be driven in the wrong direction.
- Organizational transformation.
- Lack of necessary documentation.
- Less predictability.



Waterfall Vs Agile Model



Activity – Research about “Agile”

1. Agile software development is based on _____ .
2. Give an example for agile software development methodology?
3. In agile development, it is more important to build software that meets the customers' needs today than worry about features that might be needed in the future. (True or False)

DevOps Model

DevOps is a software development method which focuses on communication, integration, and collaboration among IT professionals to enable rapid deployment of products.

DevOps = Development + Operations

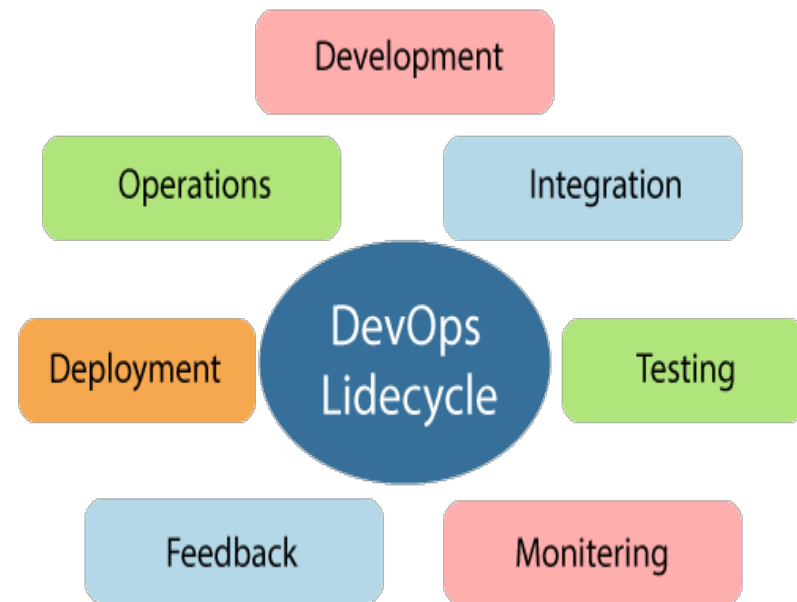
DevOps is a practice of bringing Development and Operations teams together.



DevOps Model

There are 7 stages in DevOps life cycle

1. Continuous Development
2. Continuous Integration
3. Continuous Testing
4. Continuous Monitoring
5. Continuous Feedback
6. Continuous Deployment
7. Continuous Operations



DevOps Model

Advantages

- Faster delivery of features.
- More stable operating environments.
- Improved communication and collaboration.
- More time to innovate (rather than fix/maintain).



DevOps Model

Disadvantages

- DevOps professional or expert's developers are less available.
- Developing with DevOps is expensive.
- Adopting new DevOps technology into the industries is hard to manage in short time.
- Lack of DevOps knowledge can be a problem in the continuous integration of automation projects.



Waterfall vs. Agile vs. DevOps Models

WATERFALL



AGILE



DEVOPS



Waterfall vs. Agile vs. DevOps Models

Waterfall	Agile	DevOps
Linear Approach	Continuous Iterative Sprints	Development + Operation
Comprehensive documentation	Light documentation	Light documentation
Slow delivery – major milestones (3-6 months)	Rapid delivery (daily/ weekly)	Continuous delivery
Extremely limited response to new business needs	Responsive with iterative delivery	Highly responsive with cross functional teams
Low collaboration	Improved Collaboration	High collaboration with all stakeholders
Customer feedback is infrequent / at the end of project completion	Customer feedback is frequent/ after every sprint	Customer feedback is continuous

Tools used in SDLC

Waterfall Model

- Wrike – Online project management tool that improves work visibility with its interactive Gantt charts.



- Project manager – Provides Gantt charts and task list to easily manage waterfall projects. **PROJECTMANAGER**
- Smartsheet – Online project management and collaboration software. Spreadsheet like interface.



Waterfall Model :

Smarsheet -Demo

<https://www.smartsheet.com/welcome-customers-home>

Tools used in SDLC

Agile Model

Agile tools help the team to identify the requirements and splits them into smaller tasks.

Then it tracks the programmers as they work collaboratively on the parts.

Some of the agile tools are:

- Agilo
- JIRA
- Trello
- GitHub
- Kanban



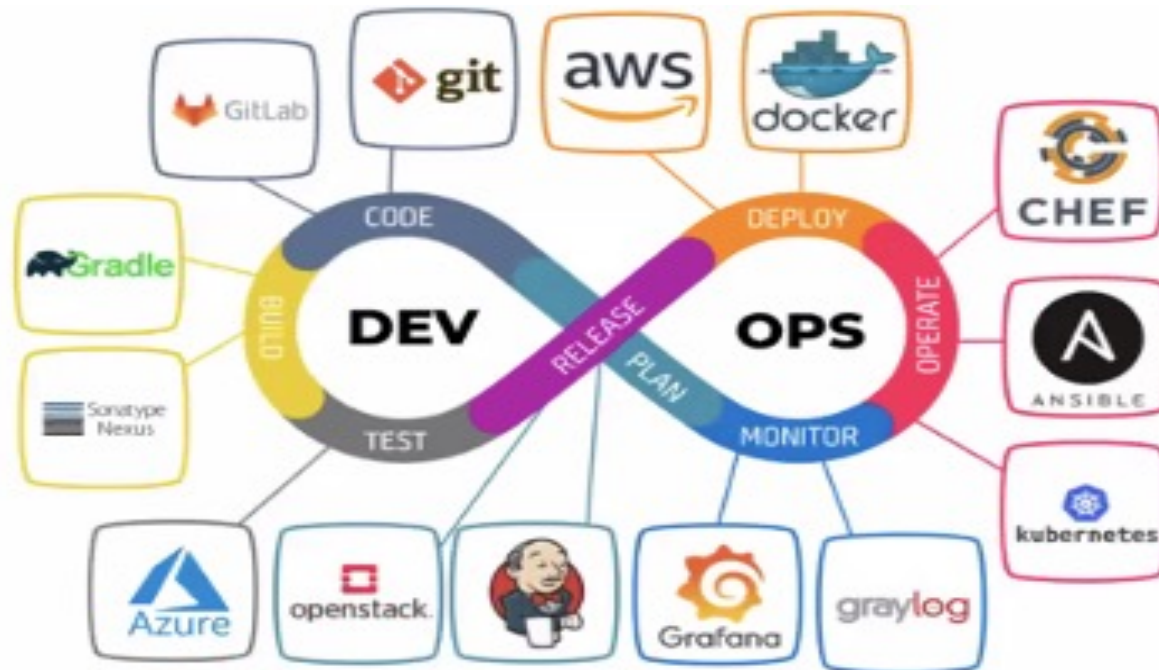
Agile Model : GitHub Demo

Creating Repositories

<https://github.com/>

Tools used in SDLC

DevOps Tools are Git, Ansible, Docker, Puppet, Jenkins, Chef, Nagios, and Kubernetes.



DevOps Model : GitLab Demo

CI / CD

<https://gitlab.com/>

Summary

1. **Agile SDLC model** is a combination of iterative and incremental process model with focus on process adaptability and customer satisfaction by rapid delivery of working software product.
2. **Agile Manifesto Principles** - Individuals & Interaction, Customer Collaboration, Responding to changes, Working Software, Working together, Simplicity.
3. Focus on **Agile – advantages & disadvantages**.
4. **Popular Agile methodologies** – Scrum, Lean, Extreme Programming etc.

Summary

5. The main **difference** is that **Waterfall** is a linear system of working that requires the team to complete each project phase before moving on to the next one while **Agile** encourages the team to work simultaneously on different phases of the project.
6. **DevOps** is a software development method which focuses on communication, integration, and collaboration among IT professionals to enables **rapid deployment of products.**
7. **Tools used in Agile-** Agilo, JIRA ,Trello, GitHub, Kanban etc.
8. Focus on **DevOps advantages & disadvantages.**